MODULE 7

Advanced Intelligence & Evidence Mapping

Certified Cyber Criminologist - Police / LEA Training

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What You Will Learn

By the end of this module you will investigate from a **safe machine**, wield **advanced tools**, and turn scattered evidence into **maps and timelines**.

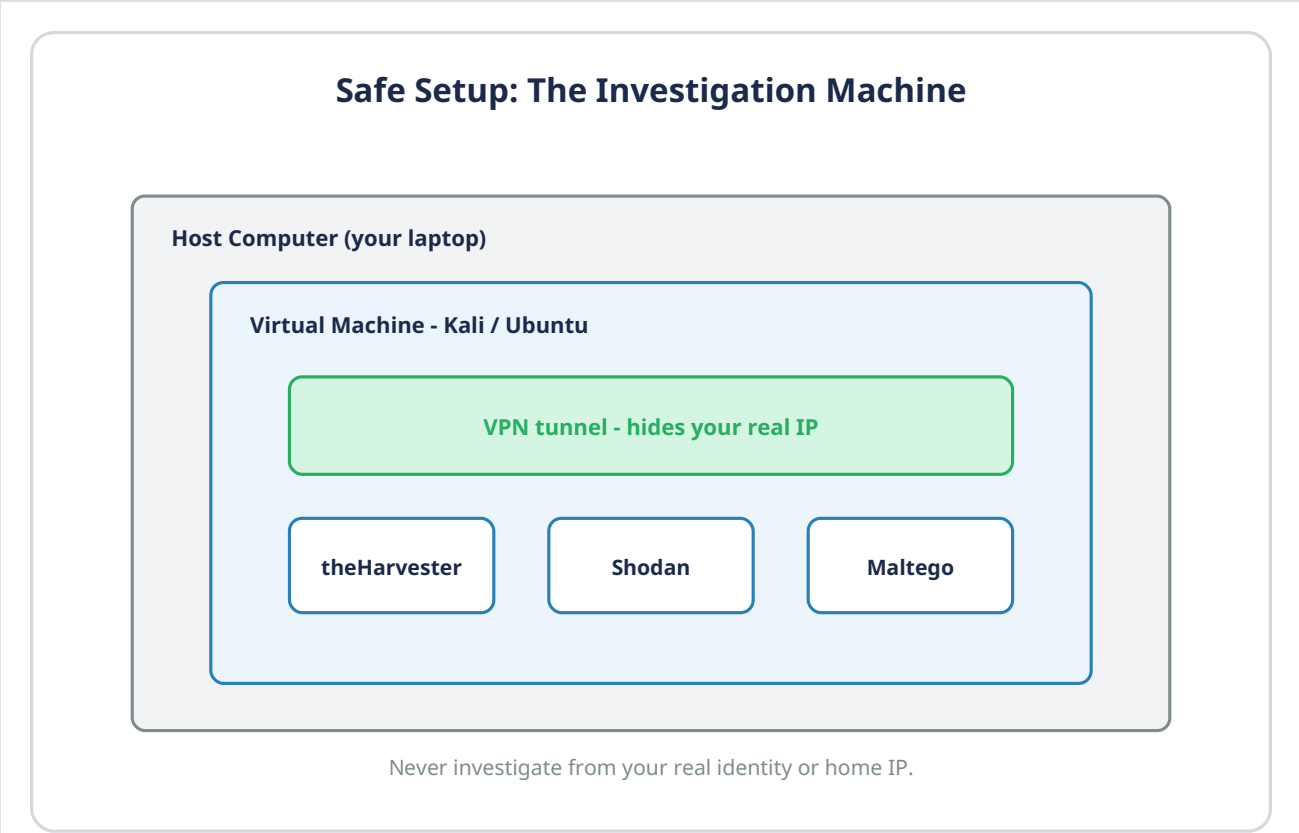
- **Standalone investigation environment** - VM, VPN, separate identity
- **Advanced OSINT tools** - Maltego, theHarvester, Shodan, Wayback Machine
- **SOCMINT** - reading criminals through their social media
- **Evidence mapping and timeline analysis** - joining the dots
- **IP to person** - the legal route, plus WHOIS and email tracing

The Case That Opens This Chapter

An investigator opens a **dangerous website** from his home computer. Next morning: a strange login attempt on **his own bank account**.

- The criminals' site had **malware** - it stole information from his machine
- New investigators make this mistake: visiting dangerous places from the **real machine**
- This chapter fixes that - **protect yourself first**, then hunt
- A compromised investigator is **worse than no investigator**

The Standalone Environment



- An **isolated system** for investigation - like a doctor's gloves with an infectious patient
- **Snapshot:** one click returns the machine to its clean state - as if nothing happened
- **Shared folders OFF** - malware must never reach your real computer

Building the Investigation VM

Your lab machine: **free tools, one afternoon** - and it will save your real computer one day.

- **VirtualBox + Ubuntu LTS:** 4 GB RAM, 25 GB disk, NAT network mode - free forever
- Take a **'clean-start' snapshot** right after setup - restore it after every investigation
- Shared folders **off or read-only**; never log into real Gmail, banking or social media inside
- Lab 1 walks you through all **16 steps** - including the common errors and fixes

Browser, VPN and Separate Identity

The VM is the room. These are the **locks on its doors**.

- **Separate browser profile:** no real logins inside - cookies and sessions stay isolated
- **VPN on:** hides your real IP; the site sees someone else - a layer, not invisibility
- **Separate email and number** for investigation work - stored in the case file when done
- Never do **personal or banking work** on the investigation machine - ever

Sock-Puppet Rules (Strict Version)

Module 6 gave you the sock puppet. Module 7 sets its **hard limits** - misuse here is a career-ending mistake.

- **View public information only** - never trap, provoke or push anyone toward a crime
- Never wear a **real person's** face or name - neutral avatar, generic details
- **Log every account:** where created, which case used it - stop using it when the case ends
- A sock puppet is a **tool for gathering**, never a toy for intrusion

Maltego and theHarvester

Two tools that turn **one clue into a network** - learn them on your own data first.

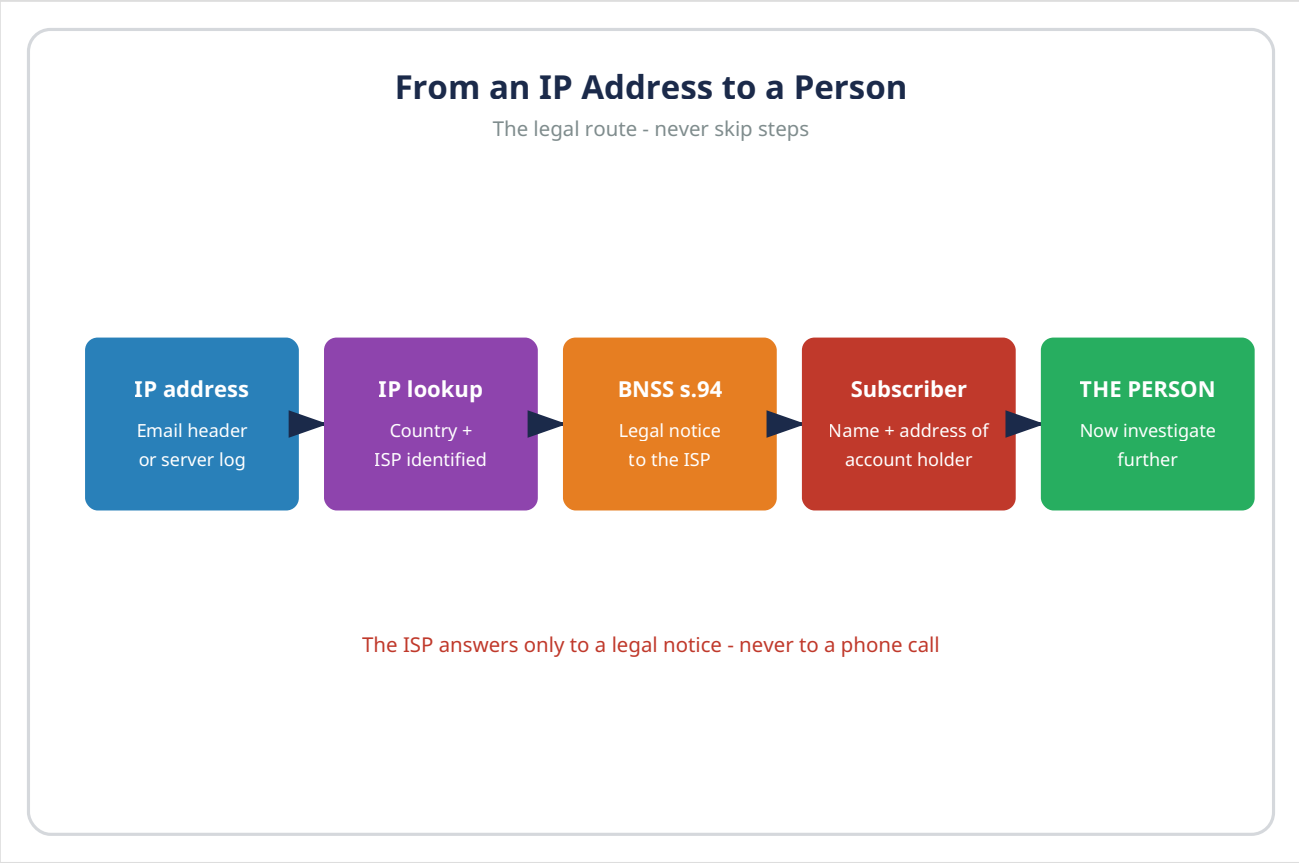
- **Maltego CE** (free): enter an email, see connected domains, IPs and accounts **as a graph**
- **theHarvester** (Kali): extracts **emails, subdomains and hosts** linked to a domain
- Command: **theHarvester -d yourdomain -b google,bing -l 200 -f result.html** - on YOUR domain only
- Watch for forgotten subdomains - **test, dev, old, staging** - attackers love them, so should defenders

Shodan, Wayback, SpiderFoot

Google finds **websites**. These tools find **everything else**.

- **Shodan:** a search engine for **devices** - open cameras, servers, routers; a free account is enough to learn
- Search **only your authorised** IP range or organisation - others' networks are off-limits without permission
- **Wayback Machine:** snapshots of changed or deleted sites - a **third party's record** for court
- **SpiderFoot:** auto-gathers from hundreds of sources - powerful but heavy; learn it gradually

From IP Address to a Person



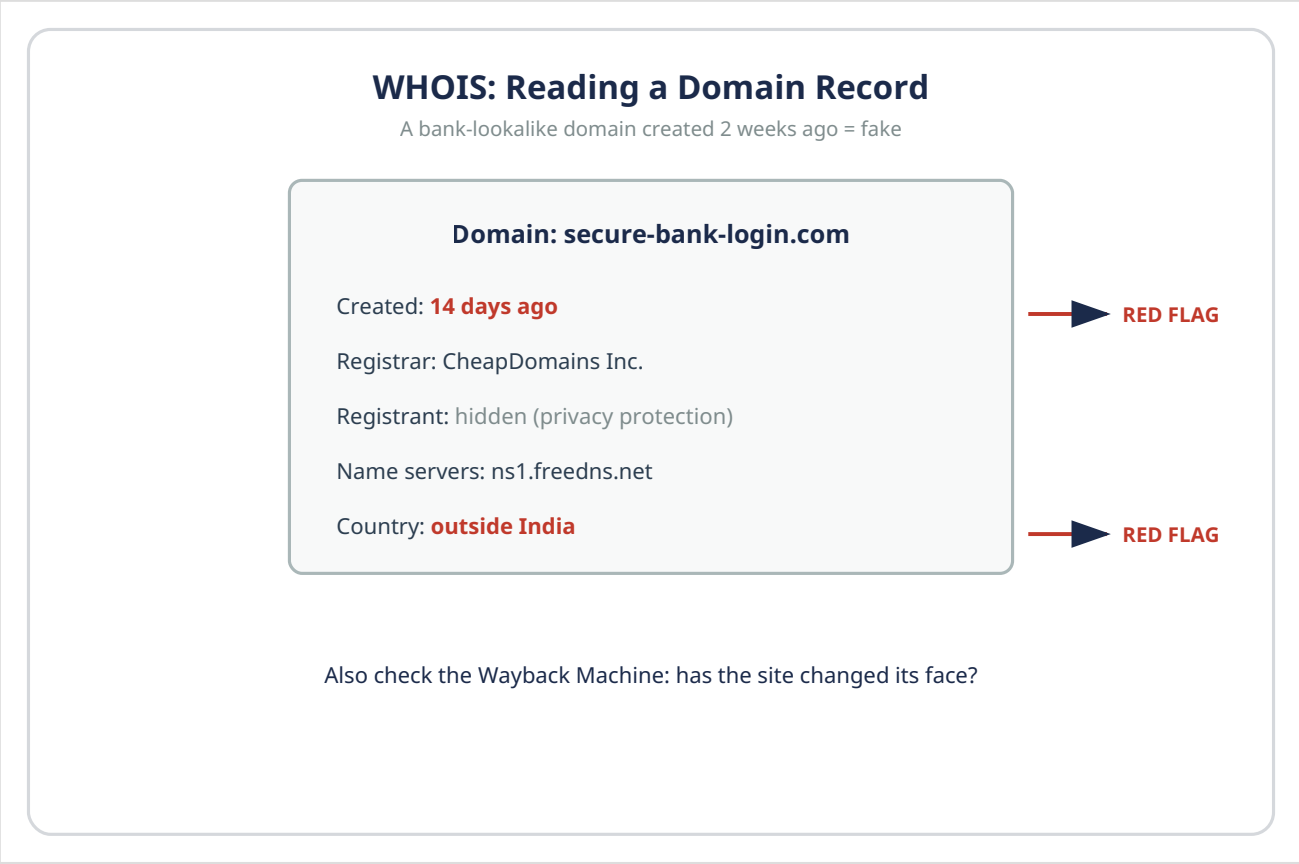
- Start from the **email header** or the **server log** - that is where the IP lives
- IP lookup gives **country and company** - not the person. The person comes next
- **Never** skip the legal notice step - shortcuts kill cases

The Legal Route: IP to Person

An IP address is **not a name**. Here is the only lawful road from one to the other.

- **Step 1:** extract the IP from the email header's **Received** line or the server log
- **Step 2:** IP lookup - identifies the **country and the ISP or company** owning it
- **Step 3: BNSS Section 94 notice** to the ISP - they reveal the subscriber behind the IP
- No notice, no name - **any shortcut makes the evidence inadmissible**

WHOIS: Reading a Domain Record



- The **creation date** is the killer detail - dates do not lie
- Registrant names are often **hidden** now - but dates and name servers are not
- Read every WHOIS record asking: **does this domain's age match its claims?**

WHOIS Red Flags

A two-minute WHOIS check **kills most fake websites** before they kill your victim's savings.

- A 'bank' website whose domain was born **two weeks ago = fake**. Real banks are decades old
- **SSL certificate check:** click the lock icon - if the certificate name and site differ, be cautious
- Recently changed name servers or **privacy-shielded everything** on a 'government' site - red flag
- Screenshot the WHOIS record with **date-time** - it is evidence

Email Tracing: Putting It Together

Module 6 taught header reading. Now use it as an **investigative weapon**.

- 'Show original' - read **Received** lines bottom-to-top to reach the true origin
- **From vs Return-Path** mismatch and failed **SPF / DKIM / DMARC** means spoofing confirmed
- Extract the IP, look it up - a **bank email from an unknown country's server = fraud**
- Practice on **your own** email first - the lab habit keeps you lawful

Browser Artifacts: The Silent Witnesses

Every browser **remembers** - history, cookies, cache, downloads. The suspect's browser is a witness that never lies.

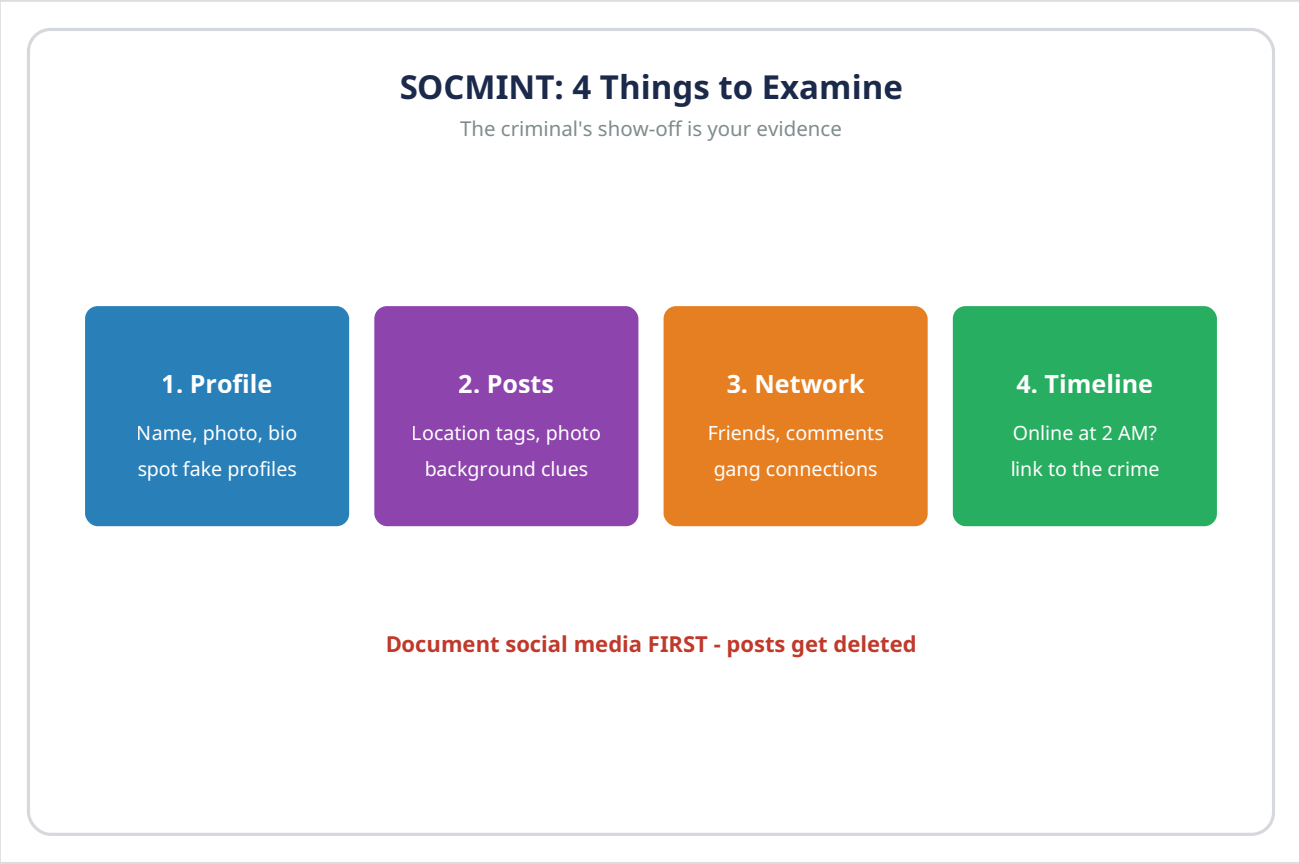
- **History:** sites visited with timestamps - rebuilds the suspect's movements
- **Cookies and cache:** prove logins and sessions - even after logout
- **Deleted does not mean gone:** forensic tools recover cleared artifacts - seize before they try
- Capture with **hash values** immediately - Module 2's chain of custody applies here

Log Basics: The Criminal's Timeline

Servers write down **everything**. Logs are the timeline the criminal did not know he was creating.

- Web, firewall and system logs record **who connected, when, from which IP**
- Merge timestamps from **all sources** - calls, transactions, logins - into one incident timeline
- One contradiction - suspect says 10, log says 11 - **opens a new direction**
- Collect logs **fast**: providers overwrite them; BNSS notices preserve them

SOCMINT: Social Media Intelligence



- Criminals **show off** fraud money - cars, phones, foreign trips
- Four lenses: **profile, posts, network, timeline** - examine all four
- One careless photo can become **a link in your chain**

Reading a Suspect's Social Media

Do not look at the face - read everything around it.

- **Fake profile signs:** new ID, few friends, stolen photo - verify with reverse image search
- **Photo backgrounds:** a shop board can reveal location; location tags confirm presence
- **Network pattern:** similar friends across profiles often means a **gang**
- **Timeline:** online at 2 AM when the crime happened at 2 AM - a link, not proof alone

Target Analysis

Deep study of the person or system in focus - **hunting the mistake every criminal makes**.

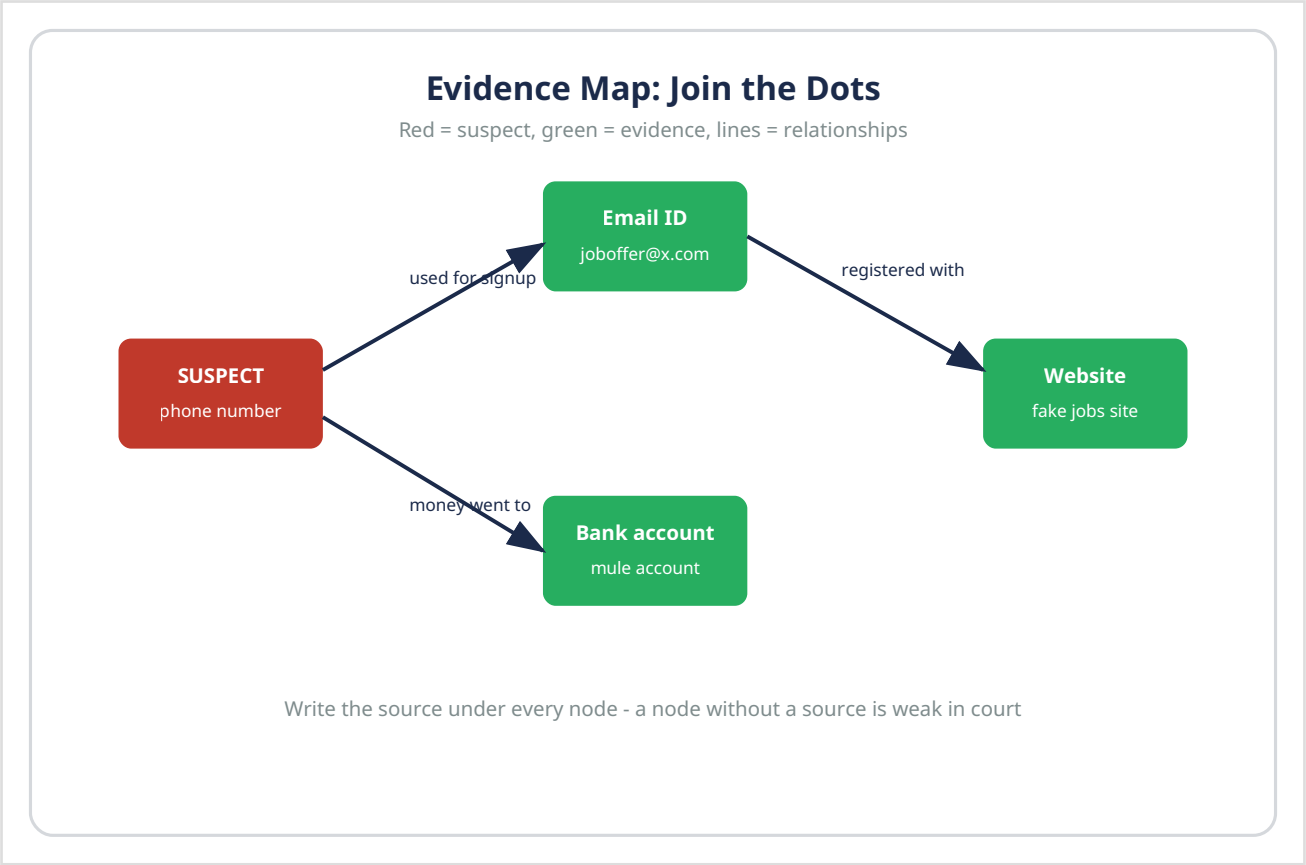
- **Three questions:** what is the target, what are its weaknesses, what are the **legal** paths to it
- Map the criminal's **online presence, methods and mistakes** - then strike at the mistake
- **Legal paths only:** hacking an account 'for investigation' is still a crime
- Document the analysis - it becomes your **investigation plan**

Sentiment Analysis

Reading **public feelings at scale** - a policing tool hiding in plain sight.

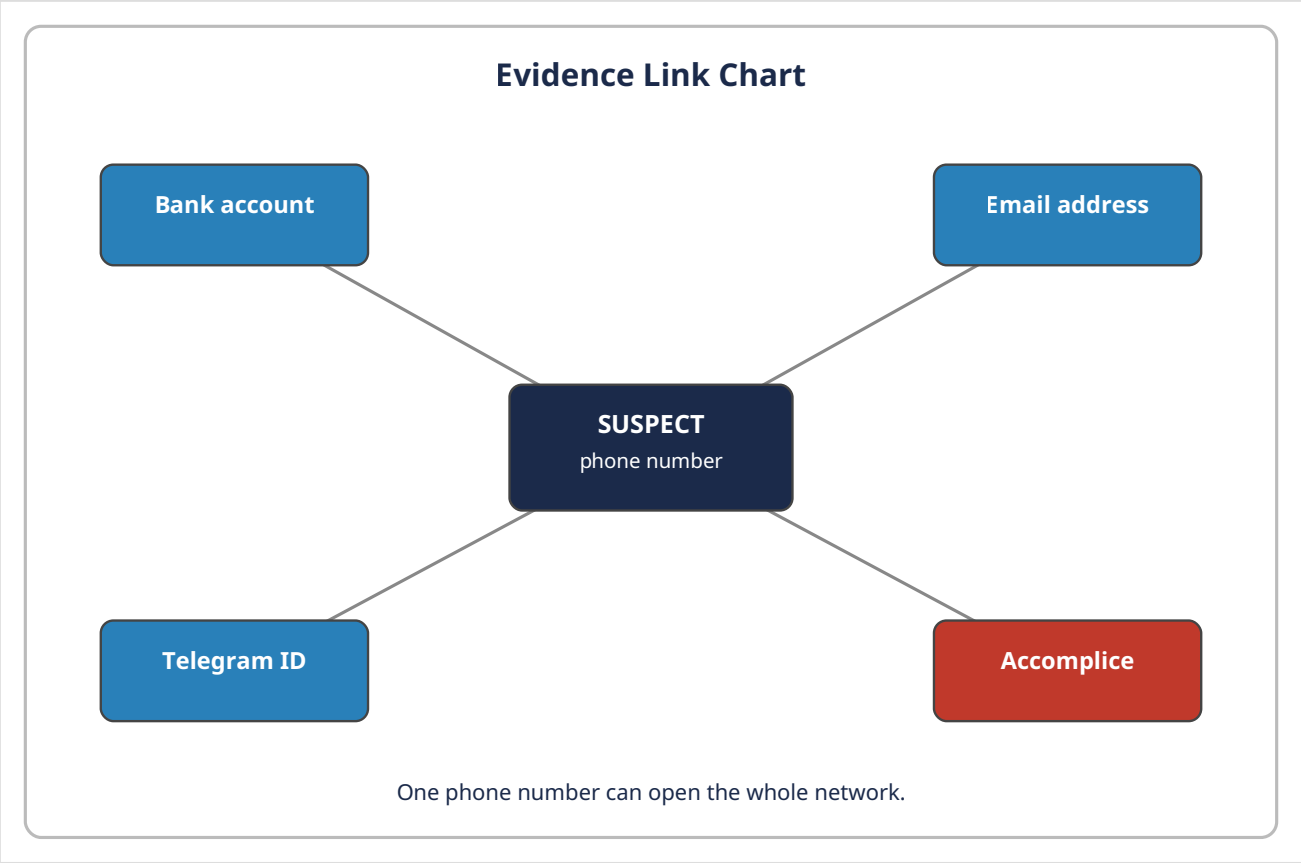
- Thousands of posts praising a 'scheme' = **fraud spreading** - greed is the fraudster's weapon
- Track **which scheme spreads where** - then warn before victims multiply
- **Distressed posts** identify victims who never filed a complaint - reach them first
- Analyse, do not surveil individuals - **patterns, not persons**

Evidence Mapping: Join the Dots



- **20 scattered pieces** - 5 numbers, 3 emails, 4 accounts - become one web
- Three rules: **label every node, label every line, source everything**
- Colours: **red** suspect, **blue** victim, **green** evidence

Evidence Link Chart (Full)



- One suspect, **many nodes** - phone, bank, email, Telegram ID - all connected
- Draw it on **paper first** - the tool is a helper; the thinking is yours
- Attach the chart to your report - **judges understand pictures**

Mapping Rules in Practice

A map without rules is **doodle**. A map with rules is **evidence**.

- Every node labelled by **type**: phone, email, account, device - use distinct shapes
- Every line labelled by **relationship**: 'same device', 'registered with', 'called 40 times'
- **Source under every node** - CDR, screenshot, theHarvester report; a node without source is garbage
- In Maltego CE, write the source in **every entity's notes** - no exceptions

Timeline Analysis

Place **every event in time order** - oldest first, each with date, time and source.

- Three gains: the **full sequence** appears, **gaps** reveal missing data, **contradictions** get caught
- Build it from **all sources**: CDR, bank records, chats, logs, Wayback snapshots
- A timeline is **court gold** - the judge sees the whole story at a glance
- Lab 4: rebuild a deleted website's timeline from the Wayback Machine

The Investigation Notebook

Your **biggest professional habit** - write every step, or lose the case twice.

- Every entry: **date, time, what was done, why, what was found, the source**
- Example: 14 Oct, 10 AM - theHarvester on college domain, 12 emails, saved result.html
- Three gains: **you never forget**, seniors get answers, **chain of custody** stays strong
- Digital notebook **plus backup** - cases run for years

Golden Rules: Protect Yourself

The investigator who gets compromised **becomes the case**. Five rules, no exceptions.

- **Real identity never exposed** - not in browsing, not in profiles, not in metadata
- **No dangerous downloads** on the real machine - the VM exists for exactly this
- No **banking or personal work** on the investigation machine - separation is safety
- **Something looks wrong? Stop.** A hasty click can make you the prey

Real Case: The Show-Off Criminal

He bought **luxury with fraud money** - then posted the proof himself.

- Expensive cars, phones, foreign trips - all over **social media**
- A **shop board** in one photo's background gave the location away
- **Location tags** confirmed presence; post **dates** built the timeline
- Lesson: **document social media first** - posts get deleted

Source: The Times of India

Preserving Social Media Evidence

A screenshot is **not** evidence until it is **preserved properly**. Do it right the first time.

- **Screenshot + URL + date-time** visible - a bare image proves nothing
- Save the page on **web.archive.org** - even if deleted later, the record survives
- **Screen recording** on phone for video content; case number and date in **every filename**
- This habit makes evidence **durable in court** - build it now

Your Turn: Labs in This Module

Five labs - **the most important part** of this chapter. Complete every one, step by step.

- **Lab 1:** build your Investigation-VM and master the clean-start snapshot
- **Lab 2:** run theHarvester on **your own** domain - see your exposure
- **Lab 3:** Shodan your **authorised** network - write a defender's report
- **Lab 4 + Lab 5:** rebuild a deleted site's timeline, then draw your first evidence-link chart

Key Takeaways

If you remember nothing else from Module 7, remember this:

- Investigate from a **separate, isolated machine** - always
- **Maltego, theHarvester, Shodan, Wayback** - master one tool at a time
- The criminal's **show-off is your evidence** - document it first
- **Maps and timelines** turn scattered data into a story a judge understands
- IP to person runs on **BNSS Section 94 notices** - never shortcuts
- Write **every step** down - the notebook is your second memory

Review Questions

- 1. Your unit's server may be exposed on Shodan. Describe your lawful check - and what you must NOT touch even with permission.
- 2. A phishing email's IP points to a foreign server. List the exact legal steps to reach the person behind it.
- 3. Draw a 5-node evidence map for a UPI fraud: label each node, each link, and the source under each node.
- 4. A suspect's alibi says he slept at 11 PM; his posts show activity at 2 AM. How do you use this - and what is it NOT?
- 5. Why must the investigation VM's shared folders stay off? Describe the attack path if they are on.

NEXT: MODULE 8

Social Media and Web Intelligence

Profiles, platforms, takedown, web leads